Zhaoxuan Wei

Institute of Operations Research and Analytics, National University of Singapore
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Education

National University of Singapore — Singapore
Ph.D. Candidate in Operations Research — Jan 2021–Present

- Advisors: Professor Hanqin Zhang and Professor Andrew E.B. Lim

Wuhan University — Wuhan, China
B.Sc. in Mathematics — 2016–2020

Research Interests

Methodologies: Stochastic Modeling, Dynamic Programming, Reinforcement Learning, Online Learning, Queueing Theory.

Applications: Decentralized Operations, Service Systems, Inventory Management, Last-Mile Delivery, Supply Chains.

Papers

Under Revision or Working Papers

1. **Multi-Agent Queueing Network: Coordination with Learning**
   with Andrew E.B. Lim and Hanqin Zhang
   2026 OR Society of Singapore Best Student Paper Award
   In preparation for submission to Operations Research

2. **Learning to Coordinate in Multi-Agent Decision Making**
   with Andrew E.B. Lim
   In preparation for submission to Operations Research

3. **Blessing of Order Crossover in Inventory Systems**
   with Yunfang Yang
   Major Revision at Operations Research

4. **Pooling vs. Proximity: Design of Parcel Locker Networks with Endogenous Service Rates**
   with Quanmeng Wang, Guodong Lyu
   Under Review at Management Science

5. **The Lyapunov-Function-Based UCB Algorithm**
   with Andrew E.B. Lim, Hanqin Zhang
   Under Review at Operations Research Letters

6. **Inventory System with Impatient Demand and Latent Backlogs**
   with Andrew E.B. Lim and Hanqin Zhang
   In preparation for submission to Management Science

Work in Progress

1. **Multi-Agent Stochastic Control and Decentralized Maximum Principle**
   with Andrew E.B. Lim

2. **Assemble-to-Order Generalized-M Systems with Stochastic Leadtimes and Lost Sales**
   with Hanqin Zhang

Teaching Experience

National University of Singapore Singapore
Instructor, DAO1704 Decision Analytics Using Spreadsheets Aug 2025–Dec 2025

- Undergraduate core module covering business analytics, spreadsheet modeling, optimization, and data analytics.
- Weekly lectures covering teaching materials

National University of Singapore Singapore
Teaching Assistant, BDC6112 Stochastic Processes 2023 August-December

- PhD core module covering Poisson and renewal processes and Markov chains.
- Weekly tutorial lectures.

National University of Singapore Singapore
Teaching Assistant, IE6001 Introduction to Optimization 2021 August-December

- PhD core module covering linear and convex optimizations.
- Weekly tutorial lectures.

Selected Talks

Multi-Agent Queueing Network: Coordination with Learning

- INFORMS Annual Meeting, 2025.
- INFORMS Applied Probability Society Conference, 2025.
- Analytics for X, 2025

Inventory Systems with Impatient Demand and Latent Backlogs

- INFORMS Annual Meeting, 2024.
- INFORMS Annual Meeting, 2023.
- INFORMS Applied Probability Society Conference, 2023.

Blessing of Order Crossover in Inventory Systems

- INFORMS Applied Probability Society Conference, 2025.

Professional Service

- Reviewer for SIAM Journal on Control and Optimization, and Naval Logistic Research.
- Session chair, INFORMS Applied Probability Society Conference, 2025.
- Session chair, INFORMS Annual Meeting, 2023

Skills

Programming: Python, MATLAB.

Languages: English, Chinese.

References

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Associate Professor Zhenyu Hu
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National University of Singapore
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